

Subhajit Chowdhury

Cloud Data Engineer

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SUMMARY

Cloud Data Engineer with **4 years of experience** architecting **robust, scalable, production-grade** ETL/ELT pipelines, near-real-time data lakehouses, and cloud migration solutions on AWS for UK/US enterprise clients. Proven track record acting as a **trusted technical advisor** to BI, QA, and analytics teams, **generating GBP 1.5M+ in annualised infrastructure cost savings** and shipping **150+ distributed pipelines** processing **hundreds of GBs daily** with **sub-minute CDC latency**. Strong cross-functional collaborator across distributed UK/US/India teams. Actively pursuing **AWS DEA-C01**.

CERTIFICATIONS

- **AWS Certified Cloud Practitioner** | Validation: a6447414ae25419cba12941481cd10a2 *July 2025 – July 2028*
- **AWS Certified Data Engineer – Associate (DEA-C01)** *In Progress – 2025*

SKILLS

- **AWS:** Glue (PySpark), EMR, S3, Redshift, Athena, Lambda, Step Functions, DMS, RDS, EventBridge, DynamoDB, QuickSight, IAM, CloudWatch
- **Languages:** Python (Pandas, NumPy), SQL, PySpark, Shell Scripting
- **Orchestration & CI/CD:** Step Functions (production), Airflow (familiar), Jenkins, Git, GitHub Actions
- **Storage & Databases:** Parquet, Delta Lake, Iceberg, Snowflake, Snowpipe, Redshift, PostgreSQL, MySQL, MS SQL Server, MongoDB, DynamoDB, Oracle
- **Patterns:** Distributed Systems, Medallion Architecture, CDC/SCD, Event-Driven Pipelines, Data Lakehouse, Cloud Migration, Cost Optimization, Data Governance, Data Observability, DataOps, Serverless, Agile/Scrum

EXPERIENCE

Tata Consultancy Services (TCS)

Kolkata, India

Data Engineer

July 2022 – Present

- **Cloud Migration & Cost Optimization:** **Transformed** two legacy on-prem client environments (MS SQL Server, MongoDB) into CDC-enabled AWS serverless lakehouses using **DMS, Lambda, Glue, Step Functions, EventBridge, S3, Athena, DynamoDB, PostgreSQL, QuickSight** – **unlocked GBP 1.5M+ annualised savings** and delivered sub-minute data latency across UK/US enterprise clients.
- **Scalable Pipeline Architecture:** **Spearheaded 150+ production ETL/ELT pipelines** on **Glue (PySpark), EMR, S3, Redshift, Athena** in medallion architecture (Bronze/Silver/Gold) – engineered **distributed systems** processing **hundreds of GBs daily** via incremental CDC and **TB-scale** full reloads under zero data-loss SLA.
- **Near-Real-Time Ingestion:** **Pioneered** event-driven ingestion using **Lambda, S3, Snowflake, and Snowpipe** – replaced scheduled batch dependencies with **sub-minute end-to-end latency**, eliminating manual data-request handoffs for downstream teams.
- **Performance & Cost Engineering:** **Accelerated Athena and Redshift** query performance by **25%** via partition pruning, predicate pushdown, and **Parquet** columnar tuning – directly cutting cloud compute costs and slashing dashboard latency for business stakeholders.
- **Cross-Functional Trusted Advisor:** **Acted as trusted technical advisor to 3+ BI, QA, and analytics teams** – institutionalised automated schema discovery, drift detection, and lineage tracking across **Parquet** and **JSON** datasets, reducing downstream data latency by **30%** and strengthening data governance.
- **CI/CD & Platform Reliability:** **Democratised** CI/CD by engineering scoped **Jenkins** self-service access for QA teams – eradicated **50+ daily engineering interruptions** and ensured high-availability platform delivery across distributed UK/US/India squads.

PROJECTS

- **Near-Real-Time Event-Driven ELT Pipeline** *Lambda, DynamoDB, S3, Snowflake, Snowpipe*
 - **Architected** a **highly-available, event-driven ELT pipeline** for continuous ingestion of live API data – **replaced a 4-hour batch process**, achieving **sub-minute latency** and eliminating **100%** of downstream data-request tickets.
 - **Engineered** serverless JSON ingestion from **S3** to **Snowflake** via **Snowpipe** – delivered same-day data availability and **robust scalability** without scheduled-run dependency.
- **Serverless Batch Analytics Lakehouse** *Glue, S3, Athena, PySpark, Step Functions*
 - **Architected** a **scalable medallion lakehouse** on **S3** with **Glue/PySpark** and **Step Functions** orchestration – delivered self-serve **Athena SQL** at **zero standing cluster cost**.
 - **Modeled** curated **Bronze/Silver/Gold Parquet** datasets – empowered analysts to run ad-hoc queries independently, optimising query performance and eliminating engineering bottlenecks.

EDUCATION

Modern Institute of Engineering and Technology | MAKAUT

2018 – 2022

B.Tech – Computer Science and Engineering (CSE)

DGPA: 8.90 / 10.0